# LLM Watermarking for Intellectual Property Protection: A Comparative Study of KGW and EXP

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## 1 Introduction

### 1.1 Research Background

With the rapid iteration and large-scale deployment of large language model (LLM) technologies, artificial intelligence generated content (AIGC) has been widely adopted in areas such as copywriting, knowledge dissemination, and business document composition. Text generated by LLMs offers advantages in high efficiency and diverse expression, but it also poses serious challenges to intellectual property (IP) protection. On one hand, the ownership of AI-generated content is ambiguous, making infringement difficult to trace. On the other hand, attackers can modify generated content through paraphrasing, synonym substitution, and text rewriting, further complicating content provenance and forensic evidence collection. Against this backdrop, watermarking techniques for LLMs have emerged as a core solution to these IP protection challenges.

### 1.2 Technical Definition and Core Evaluation Criteria

LLM watermarking technology embeds imperceptible feature signals into the text generated by the model or into the model weights themselves. Without compromising text readability, it enables traceability and authorship attribution of AI-generated content, thereby providing technical support for IP protection of LLM outputs. A practically useful LLM watermarking algorithm must simultaneously satisfy three core performance requirements:

- **Effectiveness:** The watermark detection algorithm can stably and accurately identify watermarked text.

- **Imperceptibility:** The watermark embedding process has minimal impact on text fluency and semantic integrity, remaining unnoticeable to human observation.
- **Robustness:** After common removal attacks such as paraphrasing, rewriting, or noise injection, the watermark signal remains detectable.

### 1.3 Current Research Status

The academic community and open-source community have proposed various LLM text watermarking schemes. Among them, the KGW algorithm, based on vocabulary probability distribution perturbation, and the EXP algorithm, based on exponential probability bias, are two representative classic approaches. Both have been engineered and made available via GitHub open-source projects, and are widely used for algorithm reproduction and performance validation. However, existing open-source projects mostly focus on code implementation and lack a horizontal comparative analysis of the two algorithms in terms of effectiveness, imperceptibility, and robustness. Moreover, performance varies significantly across different runtime environments and parameter configurations, making it difficult to assess their applicability in real-world IP protection scenarios.

### 1.4 Work of This Paper

To address the above issues, this paper reproduces both the KGW and EXP LLM text watermarking algorithms in full based on open-source code repositories [**?**, **?**]. We conduct a comparative evaluation around the three core metrics — effectiveness, imperceptibility, and robustness — and analyze the technical characteristics and suitable application scenarios of the two algorithms. Our work provides a reference for the engineering deployment and future optimization of LLM watermarking technologies.

### 1.5 Paper Structure

The chapters are organized as follows: Chapter 1 Introduction; Chapter 2 Problem Definition; Chapter 3 Algorithm Design; Chapter 4 Experimental Procedure; Chapter 5 Evaluation Results; Chapter 6 Conclusion.

## 2 Problem Definition

This chapter clarifies the research problem, experimental constraints, and quantitative evaluation criteria in the context of LLM text watermark deployment. This paper focuses solely on algorithm reproduction and comparative experiments, without algorithm improvements, establishing a unified framework for subsequent experimental analysis.

## 2.1 Research Questions

Existing studies mostly reproduce either KGW or EXP in isolation, lacking a horizontal comparison under identical conditions, and quantitative performance data for the three metrics are insufficient. This paper reproduces both algorithms under the same environment and conducts experiments around four specific questions:

1. Under unified conditions, what are the differences in baseline detection effectiveness between KGW and EXP, and how do key parameters affect their effectiveness?
2. Compare the text fluency after watermark embedding for both algorithms, and analyze how parameter variations affect imperceptibility.
3. Under synonym substitution and word deletion attacks, compare the robustness of the two algorithms against tampering.
4. Based on experimental data, summarize the advantages, disadvantages, and suitable deployment scenarios for each algorithm.

## 2.2 Scope Boundaries

To ensure rigor, specificity, and fair comparison in this reproduction experiment, and to avoid overgeneralization, we explicitly define the boundaries of the research subject, content, experimental scenarios, and environment.

- **Research subject:** Only text-based LLM watermarks are studied; model weights and audio-visual multimodal watermarks are excluded.
- **Research content:** Algorithms are reproduced strictly following the original papers' source code; no modifications or optimizations are made to the algorithmic structure or computational logic.
- **Test scenarios:** Robustness tests are limited to two common manual tampering methods — synonym substitution and word deletion — while extreme attacks such as model fine-tuning are excluded.
- **Experimental environment:** Both algorithms share the same software/hardware stack and baseline parameters to ensure controlled variable conditions and fair comparison.

## 2.3 Definition of Core Evaluation Metrics

To enable quantitative evaluation of algorithm performance, and in line with common practices in the LLM watermarking field and the visualization needs of this experiment, we refine the three core metrics — effectiveness, imperceptibility, and robustness — as follows.

### 2.3.1 Watermark Effectiveness

Effectiveness measures the algorithm's ability to identify watermarked text. It is the foundation for traceability and copyright determination, requiring accurate discrimination between watermarked and non-watermarked (original) text. This experiment uses two metrics: True Positive Rate (TPR) and separation of detection distributions.

**True Positive Rate (TPR).** TPR is the core quantitative metric for evaluating watermark detection capability, representing the proportion of all watermarked samples correctly identified as carrying a watermark:

$$\text{TPR} = \frac{\text{Number of watermarked texts correctly identified}}{\text{Total number of watermarked texts}} \tag{1}$$

TPR ranges from 0 to 1; the closer to 1, the higher the detection accuracy. In the experiments, TPR under no-attack conditions serves as the baseline, and we also observe how TPR changes when key parameters vary.

**Separation of Detection Distributions.** This metric assesses the ability to distinguish watermarked texts from ordinary non-watermarked texts. For KGW, the detection score ($z$-score mean $\mu$) is used as the statistical measure; for EXP, the $p$-value from statistical hypothesis testing is used. Lower overlap and larger mean difference indicate clearer watermark signal characteristics and better effectiveness.

### 2.3.2 Text Imperceptibility

Imperceptibility requires that the watermark embedding process does not compromise semantic integrity, fluency, or naturalness. This experiment uses Perplexity (PPL) as the primary quantitative metric.

**Perplexity (PPL).** Perplexity is a classic NLP metric for measuring text fluency and semantic coherence. Lower PPL indicates more grammatically sound, logically coherent, and natural text. We compute PPL for both original and watermarked text, and use the difference:

$$\Delta\text{PPL} = \text{PPL}_{\text{watermarked}} - \text{PPL}_{\text{original}} \tag{2}$$

A smaller $\Delta$PPL implies weaker negative impact and better imperceptibility.

**Text Quality Stability.** This metric evaluates how much text fluency fluctuates when adjusting core parameters. If PPL increases only slightly with stronger parameters, the algorithm better balances effectiveness and imperceptibility.

### 2.3.3 Algorithm Robustness

Robustness measures the ability to still detect the watermark signal after tampering or attacks. This experiment uses two typical attack scenarios: synonym substitution and word deletion.

**Post-Attack True Positive Rate ($\text{TPR}_{\text{attack}}$).**

$$\text{TPR}_{\text{attack}} = \frac{\text{Number of watermarked texts correctly identified after attack}}{\text{Total number of attacked watermarked texts}} \tag{3}$$

**Robustness Coefficient.** To eliminate bias caused by differences in baseline detection rates between algorithms, we introduce the robustness coefficient for fair comparison:

$$\text{Robustness Coefficient} = \frac{\text{TPR}_{\text{attack}}}{\text{TPR}_{\text{no-attack}}} \tag{4}$$

This coefficient represents the fraction of watermark detection capability retained after an attack. The closer to 1, the stronger the algorithm's robustness.

# 3 Algorithm Design

## 3.1 General Principle of LLM Text Watermarking

When a large language model generates text, it scores each candidate token based on the preceding context; higher scores are more likely to be selected. Probability perturbation watermarks exploit this property as follows:

- **Embedding:** Using a secret key known only to the watermarker, the scores of some tokens are artificially increased, guiding the model to preferentially select those tokens and leaving an invisible trace.
- **Detection:** The same key is used to reconstruct which tokens were "specially marked," and the frequency of those tokens is counted to determine whether the text contains a watermark.

## 3.2 KGW Watermark Algorithm

KGW [**?**] is the most widely used and reproduced text watermarking algorithm. Its core idea is "vocabulary splitting + score boosting."

### 3.2.1 Basic Idea

Before generating each token, the secret key is used to randomly split all candidate tokens into a green list and a red list. A uniform bias $\delta$ is added to the scores of green-list tokens, making the model more likely to choose them. The resulting text exhibits a statistical pattern: green-list tokens appear more frequently. Detection exploits this pattern to identify the watermark.

### 3.2.2 Core Procedure

**Watermark Embedding:**

1. Use the secret key to generate a random splitting rule, dividing candidate tokens into green and red lists.
2. Add a fixed bias $\delta$ to all green-list tokens; leave red-list token scores unchanged.
3. Sample the next token based on the adjusted scores.
4. Repeat the above steps to generate the entire text token by token.

**Watermark Detection:**

1. Use the same secret key to reconstruct the green-list split for each token position.
2. Count the total number of tokens in the text that belong to the green list.
3. Compute a normalized detection score ($z$-score); if it exceeds the preset threshold, the text is classified as watermarked.

### 3.2.3 Key Parameters

- **Watermark strength $\delta$:** The amount of bias added to green-list tokens. Increasing $\delta$ makes the watermark easier to detect but may affect text fluency.
- **Green-list ratio $\gamma$:** The proportion of green-list tokens among all candidates at each split. The original paper recommends $\gamma = 0.5$.
- **Detection threshold:** The critical $z$-score value for deciding whether a text is watermarked. The original paper recommends 4.0, which yields a very low false positive rate.

### 3.2.4 Flowchart

## 3.3 EXP Watermark Algorithm

EXP [**?**] is a core algorithm in the MarkLLM toolkit [**?**]. Its core idea is "global probability scaling," which does not rely on vocabulary splitting but uses statistical hypothesis testing for watermark detection.

### 3.3.1 Basic Idea

Before generating each token, the secret key is used to generate a random scaling factor. The scores of all candidate tokens are multiplied uniformly by this factor. This scaling induces a systematic shift in the entire probability distribution, creating a distinct statistical signature. Detection uses a statistical test to determine whether such a shift exists in the text.

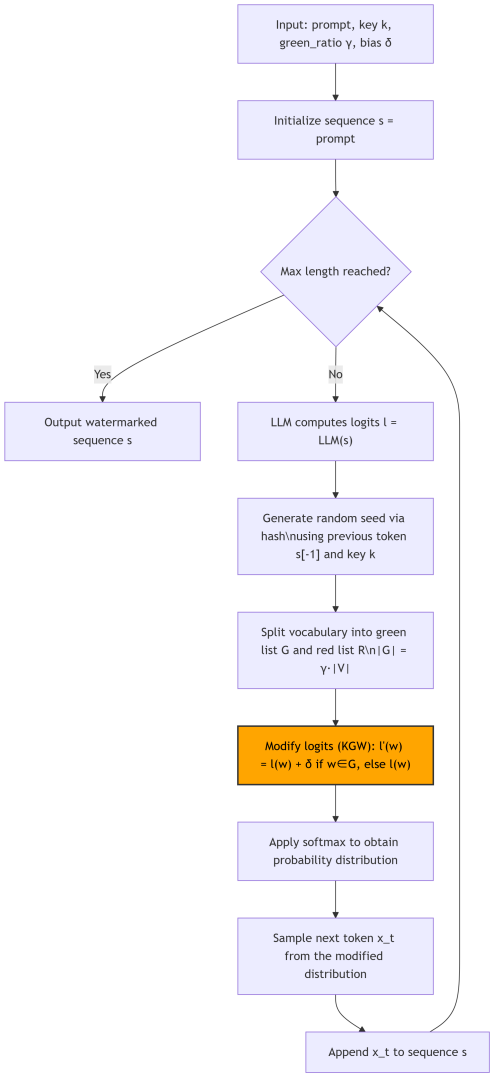



Figure 1: KGW watermark algorithm flowchart. The key step (highlighted in orange) adds bias $\delta$ to green-list token logits before sampling.

### 3.3.2 Core Procedure

**Watermark Embedding:**

1. Use the secret key to generate a random scaling factor for the current position.
2. Multiply the scores of all candidate tokens uniformly by this factor.
3. Sample the next token based on the adjusted scores.
4. Repeat the above steps to generate the entire text token by token.

**Watermark Detection:**

1. Use the same secret key to reconstruct the scaling factor for each token position.

2. Compute a statistical test statistic for the entire text.

3. Use statistical hypothesis testing to compute a significance $p$-value. The smaller the $p$-value, the higher the probability that the text is watermarked.

#### 3.3.3 Key Parameters

- **Detection threshold $\tau$:** The critical $p$-value, typically $\tau = 10^{-6}$ in our experiments. If the $p$-value is below this threshold, the text is classified as watermarked.

- **Prefix length $k$:** The number of preceding tokens used as context for the pseudorandom seed; longer prefix provides more stable seeding.

#### 3.3.4 Flowchart

### 3.4 Core Comparison of the Two Algorithms

Table 1 summarizes the key differences between KGW and EXP.

Table 1: Core comparison of KGW and EXP watermarking algorithms.

| **Aspect** | **KGW** | **EXP** |
|---|---|---|
| Core mechanism | Vocabulary splitting + green-list logit bias | Global probability scaling (no splitting) |
| Detection basis | Frequency of green-list tokens ($z$-test) | Systematic shift in probability distribution ($p$-value) |
| Implementation | Simple logic, easy to implement | More rigorous theory, slightly more complex |
| Advantages | Fast computation, low engineering cost | Clear statistical meaning, better for short texts |

### 3.5 Reproduction Notes

In this experiment, the code implementations of both KGW and EXP strictly follow the original papers without any modifications. Specifically, the EXP algorithm uses the official code from the MarkLLM open-source toolkit [**?**]. All default parameters are consistent with the original paper's recommendations, and parameter variation experiments are conducted within reasonable ranges as defined in the original papers.

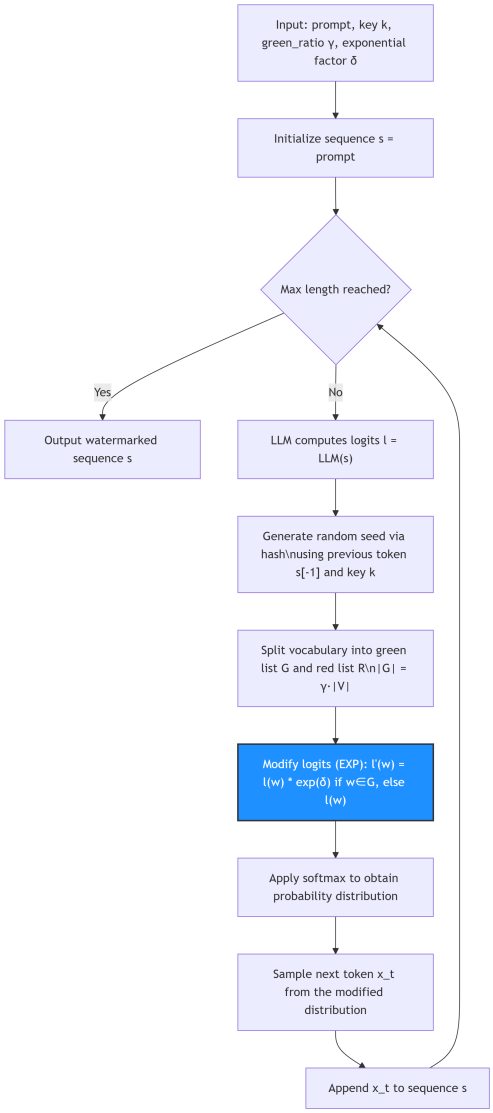



Figure 2: EXP watermark algorithm flowchart. The key step (highlighted in blue) multiplies all token logits by an exponential scaling factor before sampling.

# 4 Experiments

## 4.1 Experimental Setup

We evaluate two watermarking methods — the KGW green-list watermark [**?**] and the EXP (Exponential) unbiased watermark [**?**] — using the MarkLLM toolkit. All texts are generated by Qwen2.5-1.5B-Instruct, a publicly available small language model, with a fixed generation length of 200 tokens. The dataset consists of 20 news article prompts drawn from diverse topical domains (technology, politics, health, sports, etc.), used identically across all conditions. For each prompt, one watermarked completion and one unwatermarked (natural) completion are generated, yielding 40 samples per experimental condition. An overview of all experimental results is shown in Figure 3.

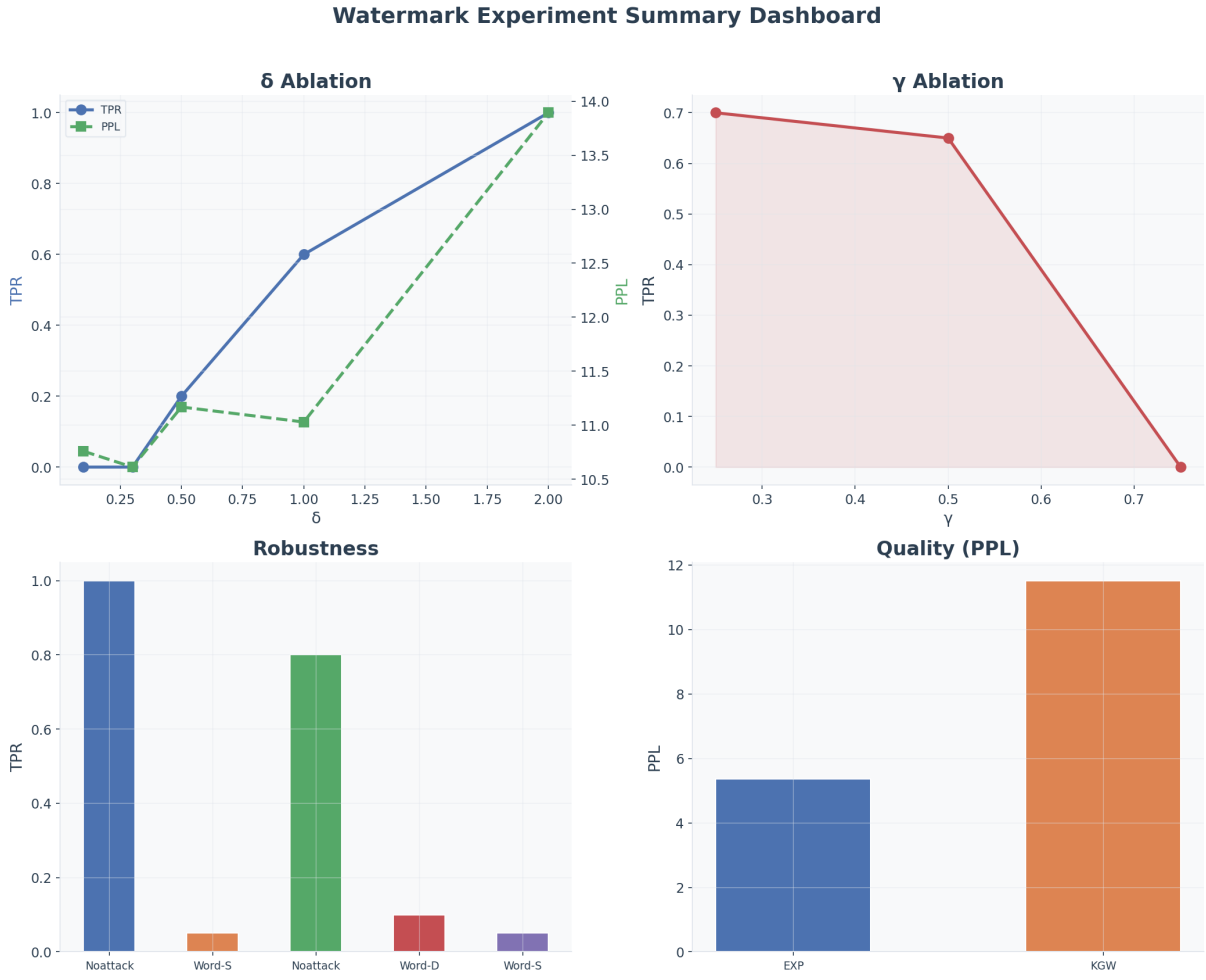



Figure 3: Summary dashboard of all watermarking experiments, showing $\delta$ ablation (top-left), $\gamma$ ablation (top-right), robustness comparison (bottom-left), and text quality comparison (bottom-right).

**KGW** partitions the vocabulary at each token step into a green list (fraction $\gamma$) and a red list, then adds a logit bias $\delta$ to green-list token scores during generation. Detection uses a one-sided $z$-test with threshold 4.0. Default configuration: $\gamma = 0.5$, $\delta = 1.0$, prefix length 1, left-window hashing scheme.

**EXP** seeds a pseudorandom function with a hash of the preceding $k$-token prefix, then resamples each token to maximize an exponentially distributed score. Detection computes an aggregate $p$-value across the sequence; a sequence is declared watermarked if $p < \tau$. Default configuration: prefix length $k = 4$, detection threshold $\tau = 10^{-6}$.

Three evaluation dimensions are assessed throughout: (1) **detection accuracy**, measured by True Positive Rate (TPR) on watermarked samples and False Positive Rate (FPR) on natural samples; (2) **text quality**, measured by perplexity (PPL) computed by the same reference model (lower is better); and (3) **robustness** under two word-level paraphrasing attacks — Word Substitution (Word-S), which replaces tokens with synonyms, and Word Deletion (Word-D), which randomly removes a proportion of tokens.

## 4.2 Detection Accuracy under Default Settings

Table 2 reports detection accuracy under default configurations ($n = 20$ per class). EXP achieves perfect detection (TPR $= 1.000$) with zero false positives (FPR $= 0.000$). KGW achieves TPR $= 0.800$ with FPR $= 0.000$. The score distributions further illustrate the separation quality. For EXP, watermarked $p$-values are vanishingly small — ranging from $10^{-103}$ to $10^{-5}$ across samples — while natural text $p$-values are distributed approximately uniformly over $[0, 1]$ as expected under the null hypothesis (Figure 5). For KGW, the mean watermarked $z$-score is $0.80$ versus $-0.43$ for natural text (Figure 4), with distributional overlap that accounts for the four missed detections at threshold 4.0.

Table 2: Detection accuracy of EXP and KGW under default parameter settings ($n = 20$ per class).

| **Method** | **TPR** | **FPR** | **Score (watermarked)** | **Score (natural)** |
|---|---|---|---|---|
| EXP | 1.000 | 0.000 | $p$-value $\approx 10^{-30}$ (mean) | $p$-value $\sim \text{Uniform}[0, 1]$ |
| KGW | 0.800 | 0.000 | $z$-score $\mu = 0.80$ | $z$-score $\mu = -0.43$ |

## 4.3 Text Quality

Table 3 compares mean perplexity of watermarked outputs against the unwatermarked baseline. Both methods produce text with lower PPL than the natural baseline (17.26), as further visualized in Figure 6. This counter-intuitive result is attributable to the fact that both mechanisms bias token selection toward higher-probability tokens: EXP's exponential resampling and KGW's positive logit shift both favor tokens that the model already assigns high probability, reducing perplexity at the cost of lexical diversity. EXP produces substantially better text quality

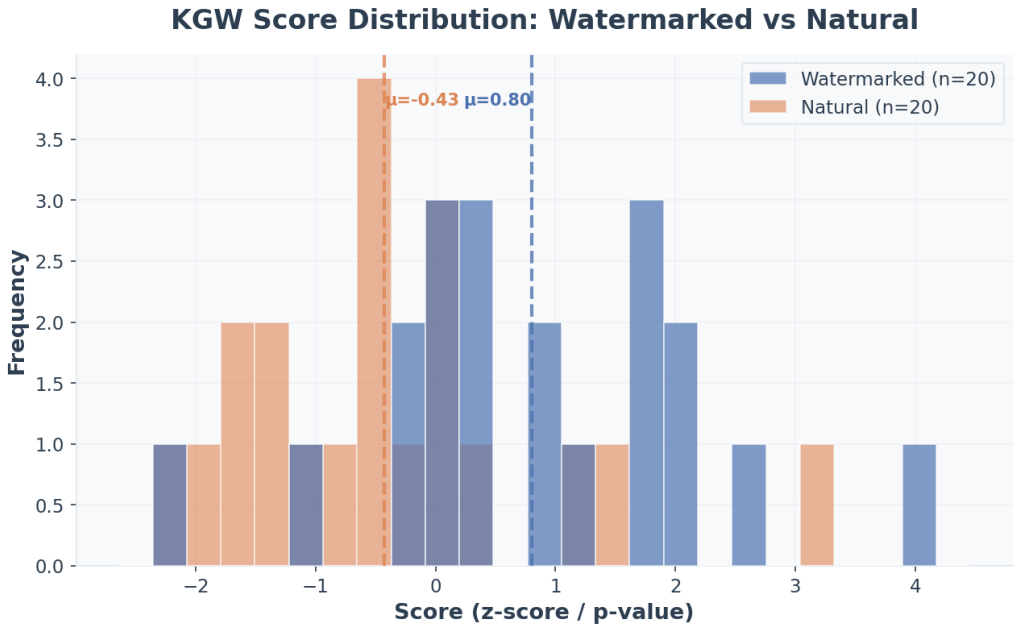



Figure 4: KGW score distributions for watermarked vs. natural text ($n = 20$ each). Watermarked mean $z$-score $\mu = 0.80$; natural mean $\mu = -0.43$.

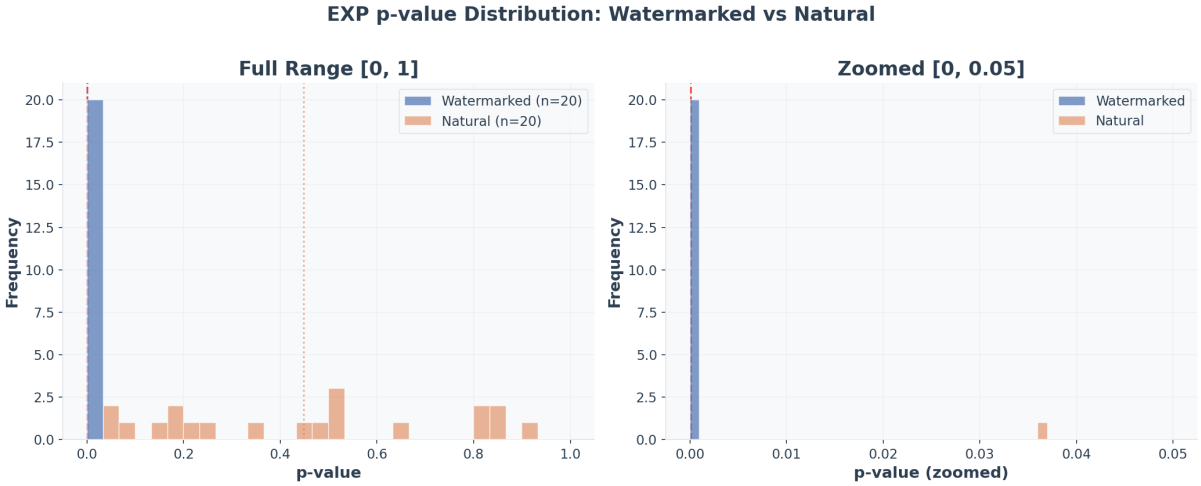



Figure 5: EXP $p$-value distributions for watermarked vs. natural text. Left: full range $[0, 1]$; right: zoomed view $[0, 0.05]$. All 20 watermarked samples cluster near zero.

than KGW (PPL 5.36 vs. 11.52), reflecting its design as an unbiased watermark whose resampling procedure preserves the relative rank of token probabilities more faithfully than KGW's additive logit shift.

Table 3: Mean perplexity (PPL) of watermarked vs. natural text. Lower PPL indicates more fluent, predictable output.

| Method | Watermarked PPL | Natural PPL | Difference |
|---|---|---|---|
| EXP (default) | 5.36 | 17.26 | −11.90 |
| KGW (default) | 11.52 | 17.26 | −5.74 |

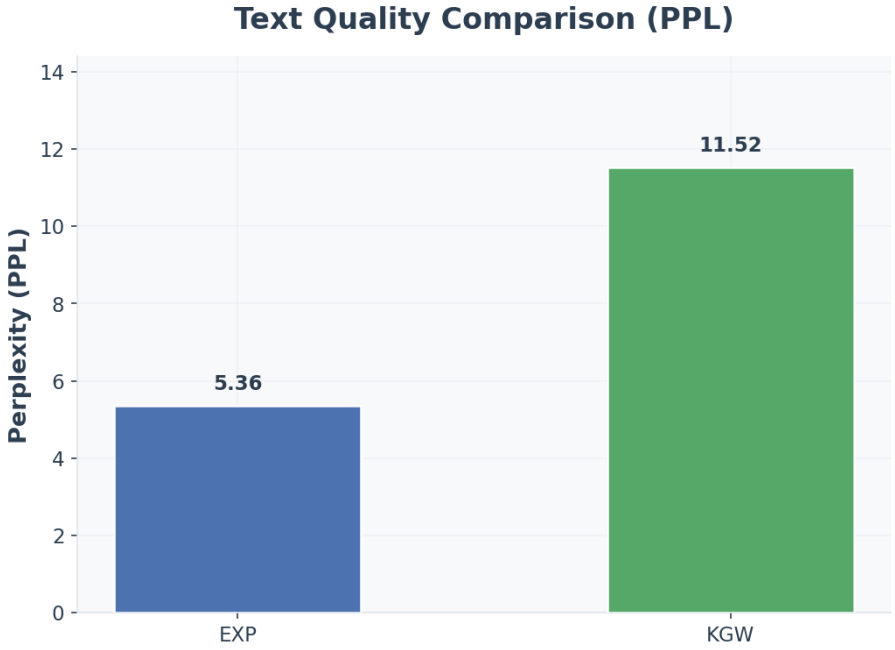



Figure 6: Text quality comparison (PPL) between EXP and KGW under default settings. Lower PPL indicates better fluency.

## 4.4 Parameter Ablation Study

### 4.4.1 KGW: Effect of Watermark Strength $\delta$

We vary the logit bias $\delta \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$ while fixing $\gamma = 0.5$. As shown in Table 4 and Figure 7, TPR increases monotonically with $\delta$: from 0.000 at $\delta \leq 0.3$, to 0.200 at $\delta = 0.5$, 0.600 at $\delta = 1.0$, and 1.000 at $\delta = 2.0$. PPL remains relatively stable for $\delta \leq 1.0$ (range 10.6–11.2) but rises sharply to 13.89 at $\delta = 2.0$, where the bias materially distorts the generation distribution. This confirms the fundamental detectability–quality trade-off in KGW: the default $\delta = 1.0$ (TPR $= 0.600$, PPL $= 11.03$) represents a moderate operating point, while achieving TPR $\geq 0.9$ requires $\delta = 2.0$ at a notable quality cost.

Table 4: KGW ablation on watermark strength $\delta$ ($\gamma = 0.5$ fixed, $n = 20$).

| $\delta$ | TPR | Watermarked PPL |
|---|---|---|
| 0.1 | 0.000 | 10.76 |
| 0.3 | 0.000 | 10.61 |
| 0.5 | 0.200 | 11.17 |
| 1.0 | 0.600 | 11.03 |
| 2.0 | 1.000 | 13.89 |

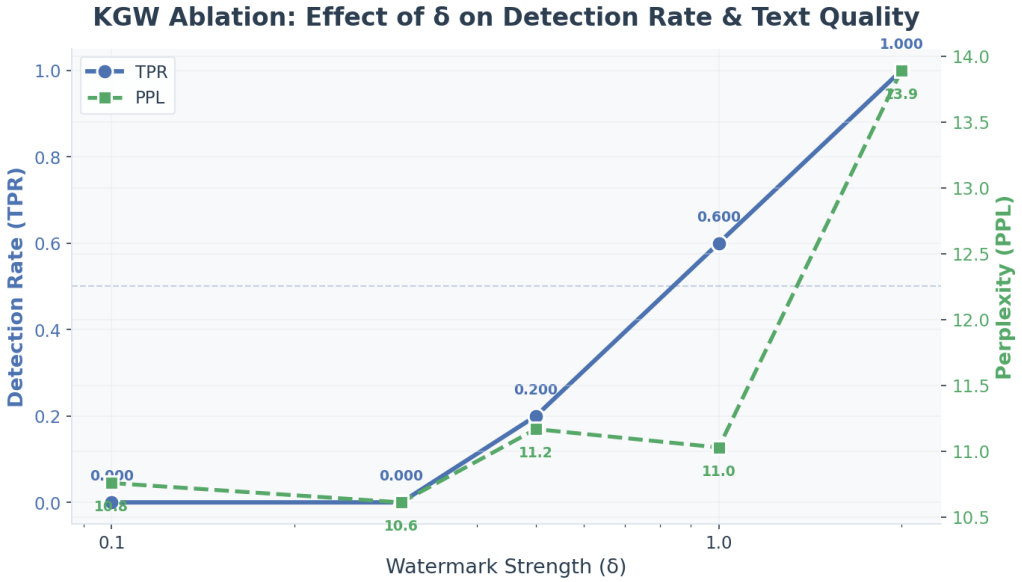



Figure 7: KGW ablation: effect of watermark strength $\delta$ on detection rate (TPR, left axis) and text perplexity (PPL, right axis). TPR increases monotonically while PPL rises sharply at $\delta = 2.0$.

### 4.4.2 KGW: Effect of Green-List Ratio $\gamma$

We vary $\gamma \in \{0.25, 0.50, 0.75\}$ while fixing $\delta = 1.0$. As shown in Table 5 and Figure 8, TPR follows a non-monotonic pattern: $\gamma = 0.50$ achieves the highest TPR (0.700), $\gamma = 0.25$ yields 0.650, and $\gamma = 0.75$ collapses to 0.000. The failure at $\gamma = 0.75$ stems from signal dilution: when nearly all vocabulary tokens are on the green list, the expected green-token proportion under the null hypothesis is close to 0.75 regardless of watermarking, so the excess $z$-score signal is negligible. PPL is stable across $\gamma$ values (11.0–12.0), confirming that the green-list ratio is not the primary driver of quality degradation.

Table 5: KGW ablation on green-list ratio $\gamma$ ($\delta = 1.0$ fixed, $n = 20$).

| $\gamma$ | TPR | Watermarked PPL |
|---|---|---|
| 0.25 | 0.650 | 11.72 |
| 0.50 | 0.700 | 11.99 |
| 0.75 | 0.000 | 10.98 |

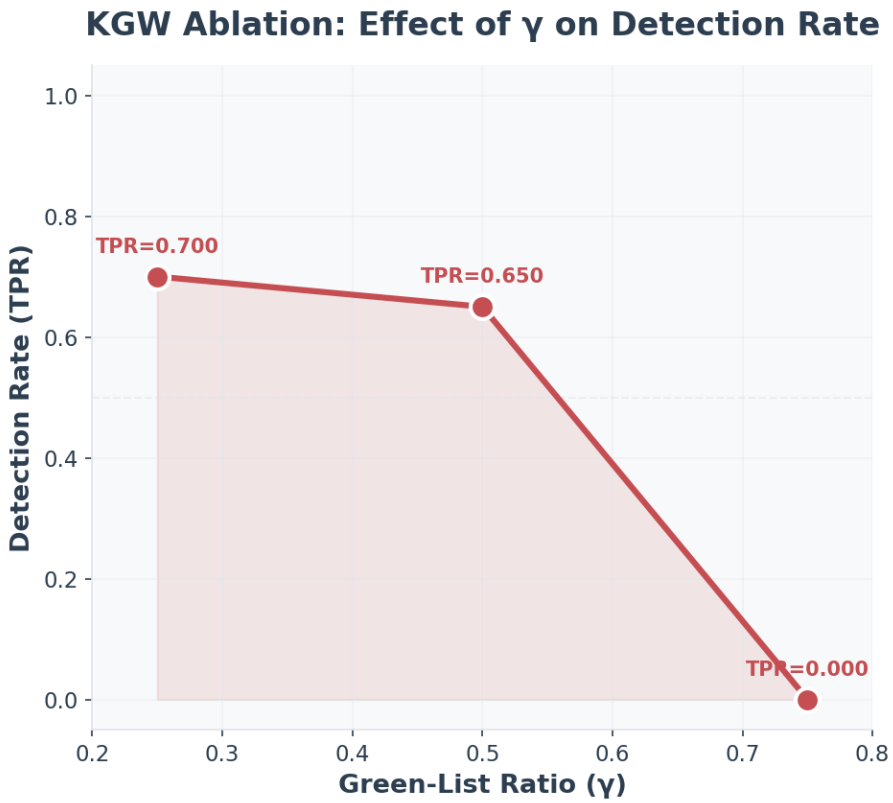



Figure 8: KGW ablation: effect of green-list ratio $\gamma$ on detection rate (TPR). TPR collapses to zero at $\gamma = 0.75$ due to signal dilution.

#### 4.4.3 EXP: Hyperparameter Sensitivity

We ablate two EXP hyperparameters independently. **Detection threshold** $\tau \in \{10^{-6}, 10^{-4}, 10^{-2}\}$ (prefix $k = 4$ fixed): TPR is 0.900 at $\tau = 10^{-6}$ and 0.950 at both $\tau = 10^{-4}$ and $10^{-2}$, with FPR $= 0.000$ throughout (Figure 9). **Prefix length** $k \in \{1, 4, 8\}$ ($\tau = 10^{-6}$ fixed): TPR is 1.000 at $k = 1$ and $k = 8$, with a minor dip to 0.950 at $k = 4$, and FPR $= 0.000$ throughout (Figure 10). In both cases, variation across settings is negligible ($\leq 0.1$ TPR). This insensitivity stems from the extremely strong watermark signal embedded by EXP: watermarked $p$-values typically lie in the range $10^{-20}$–$10^{-103}$, far below any reasonable threshold, making the exact parameter values inconsequential.

### 4.5 Robustness Against Paraphrasing Attacks

Table 6 and Figures 11–12 report TPR under two word-level attacks applied to watermarked texts before detection.

Both methods suffer dramatic degradation under attack. Word-S reduces the TPR of both methods to 0.050, effectively destroying watermark detectability in 19 out of 20 cases. Under Word-D, EXP falls to 0.000 while KGW retains a marginal TPR of 0.100.

Both collapses share the same root cause: both watermarks are fundamentally token-level

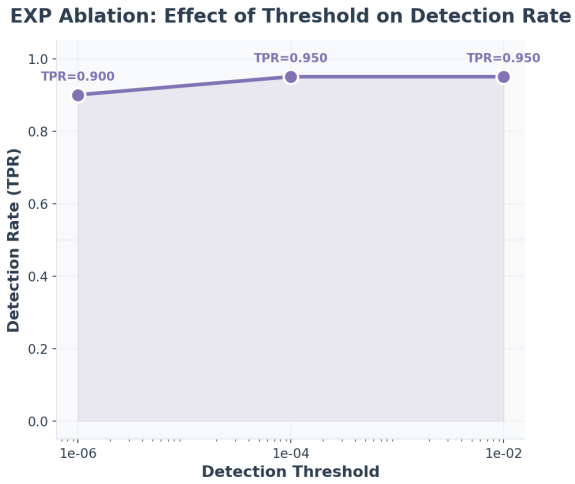



Figure 9: EXP ablation: effect of detection threshold $\tau$ on TPR. Performance is stable across three orders of magnitude.

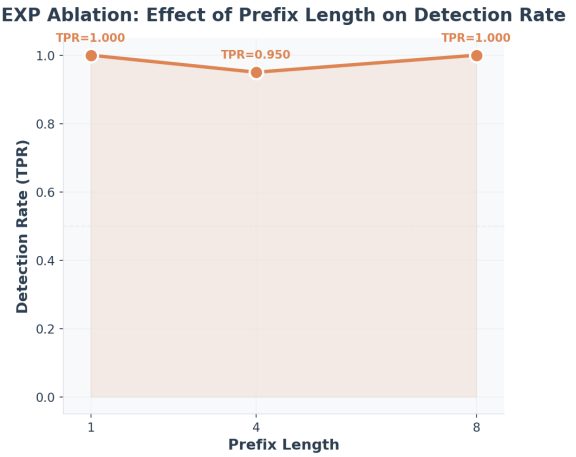



Figure 10: EXP ablation: effect of prefix length $k$ on TPR. Minor variation ($\leq 0.05$) across all tested values.

Table 6: TPR of EXP and KGW under word-level paraphrasing attacks ($n = 20$). FPR $= 0.000$ in all conditions.

| Method | No Attack | Word-S | Word-D |
|---|---|---|---|
| EXP | 1.000 | 0.050 | 0.000 |
| KGW | 0.800 | 0.050 | 0.100 |

mechanisms whose signals depend on the exact identity of generated tokens. Word-S invalidates this by changing token identities — for KGW, synonym substitution alters both the substituted token's green/red identity and the context hash for subsequent tokens, breaking the statistical signal throughout; for EXP, it disrupts the accumulated score at each modified position. The marginal advantage of KGW under Word-D (0.100 vs. 0.000) arises because deletion preserves the green/red identity of surviving tokens, allowing a small number of high-confidence samples to still exceed the $z$-score threshold, whereas EXP's joint $p$-value across the full sequence is more sensitive to the reduction in sequence length that deletion causes.

# 5 Evaluation

## 5.1 Overall Comparison

Table 7 consolidates the multi-dimensional comparison between EXP and KGW. Across nearly every axis, EXP is the stronger method: it achieves perfect detection accuracy, substantially better text quality, and comparable or equal robustness. KGW holds only a marginal advantage under Word-D attack. The one practical strength of KGW is its transparent and interpretable mechanism — the green-list partition and $z$-test are conceptually simple and easy to reason about — whereas EXP's exponential resampling is less intuitive but empirically superior.

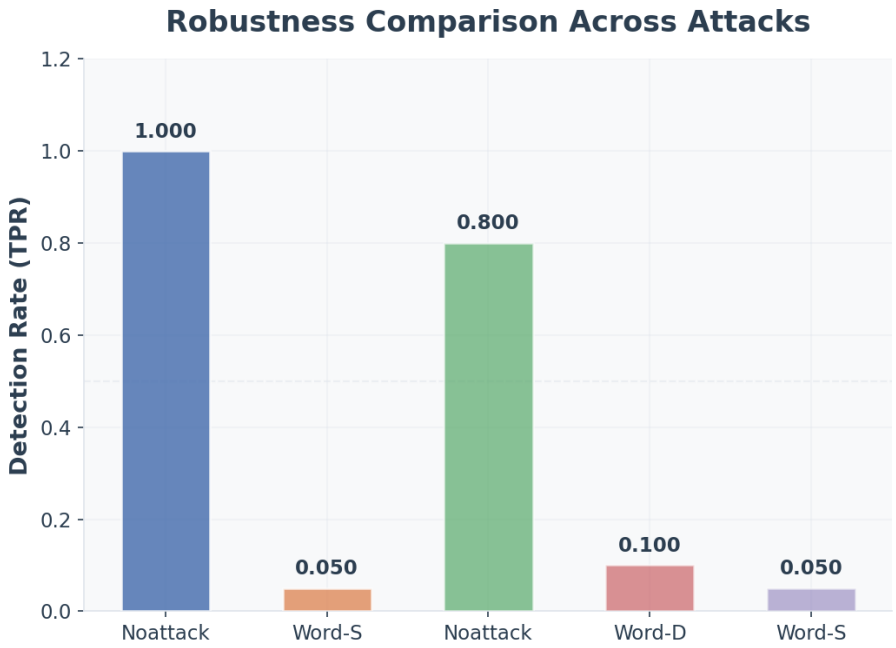



Figure 11: Robustness comparison across attack conditions. EXP (blue) and KGW (green) both collapse under Word-S and Word-D attacks.

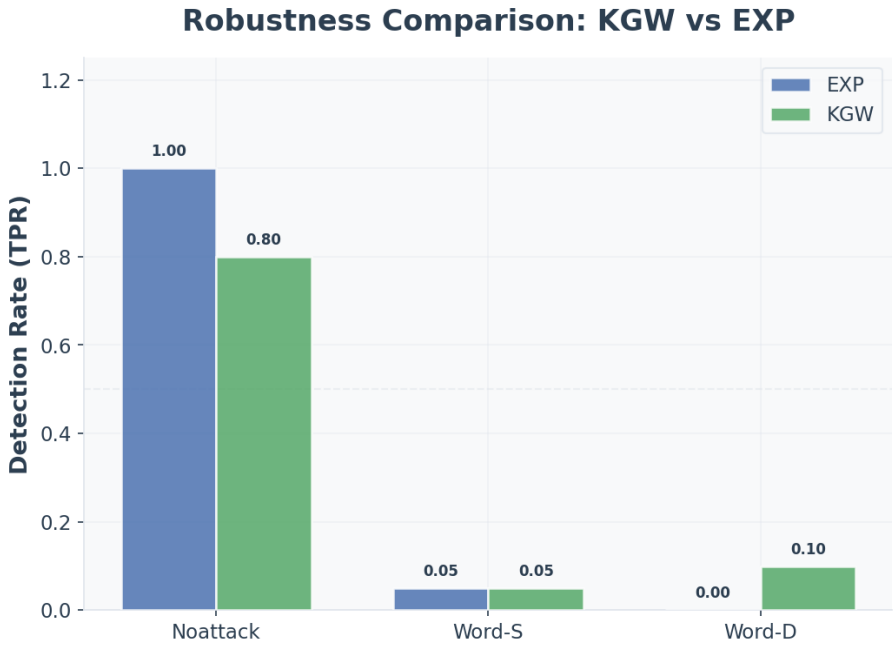



Figure 12: Side-by-side robustness comparison of EXP vs. KGW under no attack, Word-S, and Word-D conditions.

Table 7: Multi-dimensional comparison of EXP and KGW.

| **Dimension** | **EXP** | **KGW** | **Verdict** |
|---|---|---|---|
| Detection (no attack) | TPR=1.00, FPR=0.00 | TPR=0.80, FPR=0.00 | EXP |
| Text quality (PPL) | 5.36 | 11.52 | EXP |
| Robustness: Word-S | TPR=0.05 | TPR=0.05 | Tie |
| Robustness: Word-D | TPR=0.00 | TPR=0.10 | KGW (marginal) |
| Score separability | $p \approx 10^{-30}$ (wtm.) | $z$-score overlap | EXP |
| Hyperparam. stability | Low sensitivity | Sensitive to $\delta, \gamma$ | EXP |
| Interpretability | Lower | Higher | KGW |

## 5.2 Detection Accuracy

EXP demonstrates decisively superior detection over KGW in the unattacked setting. The $p$-value separation shown in Figure 5 is essentially perfect: all 20 watermarked sequences receive $p$-values below $10^{-5}$, while no natural sequence falls below $10^{-2}$. The exponential resampling mechanism ensures that each generated token contributes a statistically independent signal that accumulates multiplicatively, producing an aggregate $p$-value that grows exponentially with sequence length. At 200 tokens, this accumulation renders the signal extremely reliable.

KGW's $z$-test-based detection is inherently noisier. The $z$-score is a linear statistic that averages the green-token excess over the sequence; at 200 tokens with the default threshold of 4.0, four samples fail to exceed the detection boundary (TPR = 0.800). The score distribution in Figure 4 shows that watermarked and natural $z$-scores partially overlap, which is a structural limitation of the additive mechanism at moderate $\delta$. Raising $\delta$ to 2.0 achieves TPR = 1.000 but at a significant quality cost (PPL = 13.89), illustrating that KGW cannot simultaneously optimize both dimensions without careful tuning.

Both methods achieve FPR = 0.000 on our 20-sample natural text set, consistent with their theoretical guarantees: EXP's null $p$-value distribution is uniform by construction, and KGW's $z$-test is calibrated to a standard normal under the null. This means false positive rates can be controlled directly through the detection threshold without sacrificing sensitivity.

## 5.3 Text Quality (Stealth Analysis)

A notable finding is that both watermarked methods produce text with *lower* perplexity than the unwatermarked baseline (natural PPL = 17.26). While lower PPL superficially appears to indicate higher quality, it in fact reflects a reduction in output diversity: both methods steer generation toward high-probability tokens, producing text that is more predictable and potentially more formulaic. A sophisticated adversary comparing the statistical properties of watermarked

text against typical LLM outputs could potentially exploit this pattern as a detection signal even without access to the watermark key.

The KGW stealth analysis (Figure 13) reveals that PPL is relatively stable for $\delta \leq 1.0$ (watermarked PPL range: 10.6–11.2 across all $\delta \leq 1.0$ and all $\gamma$ settings), but rises sharply to 13.89 at $\delta = 2.0$. This identifies $\delta = 1.0$ as a practical quality ceiling for KGW: beyond this point, the logit bias distorts the generation distribution enough to produce noticeably less natural text. The $\gamma$ parameter has minimal impact on PPL (range 11.0–12.0 across $\gamma \in \{0.25, 0.50, 0.75\}$), confirming that quality degradation in KGW is driven primarily by $\delta$ rather than the green-list size.

EXP's consistently low PPL (5.36 under default settings) is its strongest practical advantage. The unbiased resampling design means that EXP generates text that, by perplexity measure, appears more fluent than even the unwatermarked baseline. This makes EXP-watermarked text harder to identify as artificially constrained, a desirable property for covert IP protection use cases.

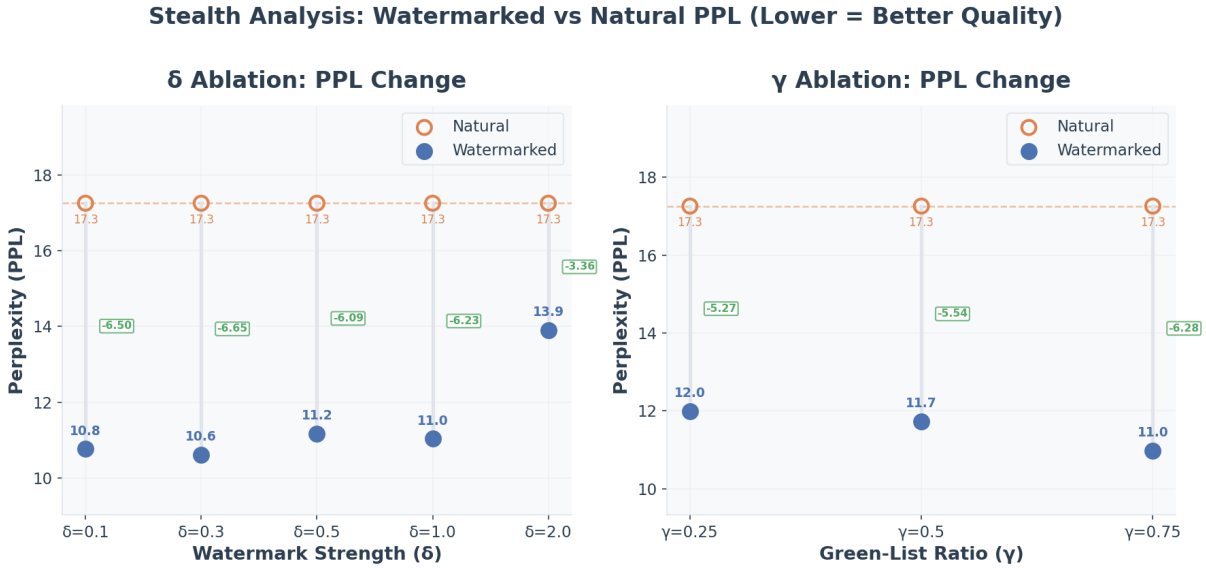



Figure 13: Stealth analysis: watermarked vs. natural PPL across KGW parameter settings. Left: $\delta$ ablation; right: $\gamma$ ablation. The orange dashed line marks the natural text baseline (PPL = 17.26).

## 5.4 Robustness

The robustness results are the most consequential finding of our evaluation: both methods fail catastrophically under synonym substitution (Word-S), with TPR dropping to 0.050 for both. This near-complete failure reveals that neither method provides meaningful intellectual property protection against an adversary who applies even a basic paraphrasing tool.

The failure is a direct consequence of the token-level granularity of both watermarks. Both EXP and KGW embed their signals through the precise identity of generated tokens. Any op-

eration that changes token identities — even while preserving semantic meaning — effectively erases the embedded signal. This is a structural limitation of the entire class of generation-time token-level watermarks, not a deficiency specific to KGW or EXP. Sentence-level or semantic-level watermarking approaches, which encode information in higher-level properties such as syntactic choices or discourse structure, would be required to withstand such attacks.

Under Word-D, KGW's marginal advantage (TPR = 0.100 vs. 0.000 for EXP) is practically insignificant — both methods fail in 18–20 out of 20 cases. The difference arises from the different detection statistics: KGW's $z$-score is computed over surviving tokens and can still exceed the threshold if enough high-signal tokens remain, while EXP's joint $p$-value degrades more rapidly with the loss of scored positions. Neither advantage translates to a robust practical defense.

### 5.5 Pros, Cons, and Deployment Recommendations

Based on our experiments, we summarize the practical deployment profile of each method.

**EXP** is the preferred choice when detection reliability and text quality are the primary concerns and the threat model does not include active paraphrasing. Its key advantages are: (1) perfect TPR and FPR = 0 at default settings; (2) significantly lower PPL (5.36), producing more natural-looking output; and (3) near-zero sensitivity to hyperparameters, simplifying deployment. Its main limitation is fragility to token-level perturbation, shared equally with KGW.

**KGW** may be preferred in settings where interpretability and parameter control are important. Its $\delta$–$\gamma$ parameter space offers an explicit, predictable trade-off between detection power and text quality, useful for practitioners who need to calibrate the watermark to specific quality constraints. However, achieving competitive TPR requires $\delta \geq 1.0$, which already causes meaningful PPL elevation (11.52 vs. EXP's 5.36). The catastrophic failure at $\gamma = 0.75$ (TPR = 0.000) is an important practical warning: the green-list ratio must be chosen carefully to avoid signal dilution.

Neither method is suitable for adversarial deployment scenarios where the watermarked text may be paraphrased. Future improvements should target robustness as the primary desideratum, for example by designing watermarks that operate at the sentence or paragraph level, or by embedding signals in semantic choices that are preserved under paraphrase.

## 6 Conclusion

In this paper, we reproduced and systematically evaluated two representative LLM watermarking methods — KGW and EXP — on text generated by the Qwen2.5-1.5B-Instruct model using the MarkLLM toolkit. Our evaluation covered three core desiderata of watermarking: detection accuracy (TPR and FPR), text quality (perplexity), and robustness against word-level paraphrasing attacks, complemented by parameter ablation studies for both methods.

Our experiments yield three principal findings. First, EXP is a strictly superior watermarking method compared to KGW under benign conditions. EXP achieves perfect detection (TPR = 1.000, FPR = 0.000) with a mean watermarked perplexity of 5.36, versus KGW's TPR = 0.800 and PPL = 11.52. The exponential resampling mechanism of EXP produces an extremely strong and consistent signal — watermarked $p$-values as low as $10^{-103}$ — while simultaneously preserving text fluency better than KGW's additive logit bias. KGW's interpretable green-list mechanism comes at a real cost: its detection reliability depends heavily on the choice of $\delta$ and $\gamma$, and misconfiguration (e.g., $\gamma = 0.75$) can cause complete detection failure.

Second, both methods are highly vulnerable to paraphrasing attacks. The Word-S (synonym substitution) attack reduces the TPR of both methods to 0.050, and Word-D (token deletion) reduces EXP to 0.000 and KGW to 0.100. These results expose a fundamental limitation of generation-time token-level watermarks: the embedded signal depends entirely on the exact identity of generated tokens, and any operation that alters token identities — even while preserving meaning — is sufficient to destroy detectability. This finding has important implications for real-world deployment: neither method provides reliable IP protection against an adversary equipped with a basic synonym substitution tool.

Third, both watermarking methods produce text with lower perplexity than the unwatermarked baseline (natural PPL = 17.26). While this might appear to be a quality benefit, it reflects a systematic bias toward high-probability tokens that reduces lexical diversity. This over-predictability is itself a potential vulnerability: a statistical test on output diversity could potentially identify watermarked text without any knowledge of the watermark key.

From this project, we learned that the key challenge in LLM watermarking is not *detection* — both methods achieve zero false positives and reasonable true positive rates under ideal conditions — but *robustness*. A watermark that is easily removed by paraphrasing offers limited practical value for IP protection. The results suggest that progress in this field requires moving beyond token-level signal embedding toward approaches that operate at a semantic or structural level, encoding information in properties of text that are inherently preserved under meaning-preserving transformations.